OBLIG 2

Analystisk utledning

$$f(x) = e^{\frac{-x}{4}} \cdot \tan^{-1} x$$

Product Regel

$$u = e^{\frac{-x}{4}}, \quad u' = -\frac{1}{4} e^{\frac{-x}{4}}$$

$$v = \tan^{-1} x, \quad v' = \frac{1}{x^2+1}$$

$$f'(x) = -\frac{1}{4} e^{\frac{-x}{4}} \cdot \tan^{-1} x + e^{\frac{-x}{4}} \cdot \frac{1}{x^2+1} \quad \Big| \; e^{-\frac{x}{4}}$$

$$f'(x) = e^{\frac{-x}{4}} \left(-\frac{1}{4} \tan^{-1} x + \frac{1}{x^2+1} \right) \quad \Big| \; e^{-\frac{x}{4}}$$

$$f'(x) = -\frac{1}{4} \tan^{-1} x + \frac{1}{x^2+1} \quad \Big| \cdot 4$$

$$\underline{\underline{f'(x) = \tan^{-1} x + \frac{4}{x^2+1}}}$$